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Stock market volatility predictability in a data-rich world: A new insight

Feng Ma [a,b], Jiqian Wang [a,b,*], M.I.M. Wahab [c], Yuanhui Ma [d]

[a] School of Economics and Management, Southwest Jiaotong University, Chengdu, China
[b] Service Science and Innovation Key Laboratory of Sichuan Province, China
[c] Department of Mechanical and Industrial Engineering, Ryerson University, Toronto, Canada
[d] School of Mathematics, Southwest Jiaotong University, Chengdu, China

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ABSTRACT

This study develops a shrinkage method, LASSO with a Markov regime-switching model (MRS-LASSO), to predict US stock market volatility. A set of 17 well-known macroeconomic and financial factors are used. The out-of-sample results reveal that the MRS-LASSO model yields statistically and economically significant volatility predictions. We further investigate the predictability of MRS-LASSO with respect to different market conditions, business cycles, and variable selection. Three factors (equity market returns, a short-term reversal factor, and a consumer sentiment index) are the most frequent predictors. To investigate the practical implications, we construct the expected variance risk premium (VRP) by using volatility forecasts generated from the LASSO and MRS-LASSO models to forecast future stock returns and find that those models are also powerful.

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1. Introduction

Stock volatility, a proxy for risk, is closely tied to risk management and asset pricing, among other things (Bauwens et al., 2012; Bollerslev et al., 2018). Accurately forecasting volatility remains an exciting task for researchers, investors, and policymakers. In light of this, many studies have improved the accuracy of volatility forecasts by constructing forecasting models or finding new predictors (Audrino et al., 2020; Barunik et al., 2016; Bekaert & Hoerova, 2014; Fang et al., 2020; He et al., 2021; Sévi, 2014; Takaishi, 2018; Tang et al., 2021; Wen et al., 2016; Wilms et al., 2021; Xiao et al., 2021; Yao et al., 2019). This study proposes a new MRS-LASSO model incorporating the Markov regime-switching (MRS) technique and variable selection methodology to forecast US stock market volatility.

This research employs realized volatility (RV) to measure stock volatility. RV contains less market noise and performs better than other measures (Andersen & Bollerslev, 1998; Audrino et al., 2020; Barunik et al., 2016; Degiannakis & Filis, 2017; He et al., 2021; Sévi, 2014; Tang et al., 2021; Xiao et al., 2021). Most studies extend the HAR-RV model of Corsi (2009) to explore whether exogenous variables can predict future stock volatility (Barunik et al., 2016; Engle et al., 2013; Paye, 2012; Sévi, 2014; Wang et al., 2018). Such studies focus on different predictors, such as the sentiment index (e.g., Audrino et al., 2020), economic policy uncertainty index (e.g., Wang et al., 2020), and CBOE volatility index (e.g., Fernandes et al., 2014). For example, Paye (2012) explores whether macroeconomic variables can improve the accuracy of volatility forecasts. With the increasing number of predictors found in volatility forecasting, how do investors and researchers extract useful information from such a data-rich environment? Additionally, from a statistical perspective, a model cannot include all the predictors recorded in previous studies due to the overfitting issue.

* Corresponding author at: School of Economics and Management, Southwest Jiaotong University, Chengdu, China.
E-mail addresses: mafeng2016@swjtu.edu.cn (F. Ma), wangjiqian@my.swjtu.edu.cn (J. Wang).

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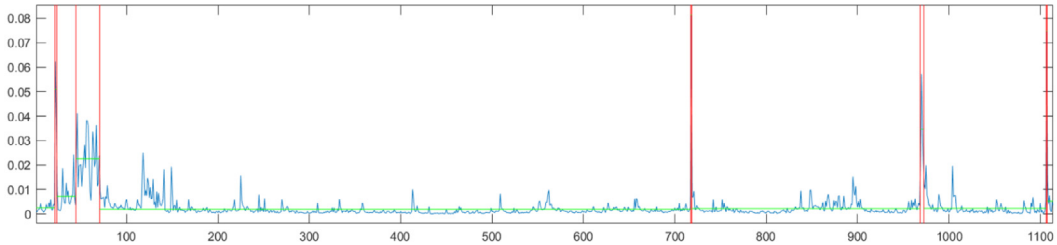

Fig. 1. Structural break test.

Thus, the prevailing shrinkage methods are introduced to address the above two questions and are widely used to select predictive factors (Cheng et al., 2021; Fang et al., 2020; Yao et al., 2019; Zhang et al., 2019a, 2019b). Therefore, we construct a novel model incorporating the MRS technique and the shrinkage method to forecast stock market volatility based on the literature.

Undoubtedly, the shrinkage method can improve the accuracy of volatility forecasts due to the bias–variance trade-off (Tibshirani, 1996; Zhang et al., 2019a). To explain why we extend the existing shrinkage method with the regime-switching technique, we provide three motivations. First, according to the arguments of McAleer and Medeiros (2008), Ma et al. (2017), and others, volatility dynamics have nonlinear features, including clustering, long memory, and structural breaks. Therefore, extending the forecasting model with the MRS technique is appropriate to accommodate nonlinearities.[1] Second, the forecasting model with MRS allows volatility to change or switch between different regimes due to factors such as extreme events and business cycles (e.g., Hamilton & Susmel, 1994; Ma et al., 2017). Hence, to obtain higher forecasting accuracy, we introduce the MRS technique. Finally, we employ the structural break test of Lavielle and Teyssiere (2007) to explore whether structural breakpoints exist in the S&P 500 monthly RV from January 1928 to November 2020. With the results as shown in Fig. 1, we can confirm structural breakpoints in the US stock market. Therefore, the MRS technique must be included in the forecasting model.

The purpose of our study is to explore whether the forecasting model incorporating MRS and variable selection exhibits superior performance in forecasting monthly RV. Several considerations motivate us to employ low-frequency (daily) data to construct the monthly RV. First, rational investors are not willing to trade frequently in the stock market because of transaction fees. Second, Lyócsa et al. (2021) argue that low-frequency volatility forecasting models can outperform high-frequency models with respect to asset allocation. Finally, most macroeconomic variables are available at a monthly or quarterly frequency. Additionally, we consider many predictors, including 12 macroeconomic and 5 financial variables, which are included in Conrad and Kleen (2020) and Fang et al. (2020). Subsequently, we construct several classical forecasting models and compare the predictive power of the MRS-LASSO model with that of other models.

The empirical design is as follows. First, based on Paye (2012), our benchmark model is the autoregressive model (AR), which is widely used in forecasting stock volatility at a monthly frequency (Wang et al., 2018). Moreover, we extend the AR model with the individual macroeconomic and financial predictors included in this study. Additionally, Rapach et al. (2010) and Zhang et al. (2019a) state that individual predictors are unlikely to maintain powerful predictability during the long term or changing market conditions due to uncertainty. Hence, 5 popular methods are introduced to generate volatility forecasts: the mean combination method, median combination method, trimmed mean combination method, and discount mean squared forecasting error combination method with two values of the discount factor. Subsequently, based on the excellent shrinkage method, we include the MRS technique and LASSO of Tibshirani (1996) and construct the MRS-LASSO model to forecast future stock volatility. Second, we generate out-of-sample volatility forecasts by the expanding (recursive) window method. Based on Paye (2012), Degiannakis and Filis (2017), Wang et al. (2018), and Tang et al. (2021), two statistical methods (out-of-sample R^2 and direction-of-change (DoC) test) and an economic methodology are used to evaluate the forecasting performance of each model. Furthermore, we focus on which model can outperform competing models and explore the forecasting performance during different market conditions and over the business cycle. Third, since the shrinkage methodology can set the coefficient of some variables to zero, we further investigate how many variables with nonzero coefficients are selected by the LASSO and MRS-LASSO models in the out-of-sample period. Fourth, as an application, accurate volatility forecasts are used to construct the variance risk premium (VRP), a powerful predictor of future stock returns (Bollerslev et al., 2009; Drechsler & Yaron, 2011). Therefore, we construct the VRP by using volatility forecasts generated by the LASSO and MRS-LASSO models and predict future stock returns to compare the out-of-sample performance. Finally, several robustness checks are conducted to reconfirm our findings, including alternative evaluation methods, subsample analysis, alternative forecasting

[1] Reviewing the existing studies, we can roughly classify the nonlinear volatility forecasting models into two groups. Some studies construct the MRS-GARCH model (GARCH model with regime-switching) to fit the nonlinear volatility dynamic (Klaassen, 2002; Lux & Kaizoji, 2007; Marcucci, 2005; Zhang et al., 2019c). The others extend the popular HAR-RV model with regime-switching techniques to improve the accuracy of volatility forecasting in the financial market (Ma et al., 2017).

horizons, alternative financial markets, and the rolling window method.

Several interesting findings are highlighted below.

(1) The out-of-sample evaluation results indicate that the forecasting model with the regime-switching technique and variable selection can consistently outperform competing models in terms of statistical and economic significance. Remarkably, the MRS-LASSO model can successfully predict stock volatility during crisis periods, including the 1987 American stock crisis, the 2008 global financial crisis, and the 2020 financial turmoil caused by the COVID-19 pandemic. Additionally, the MRS-LASSO model is more powerful when the economy becomes prosperous during the expansion period.

(2) Three variables, namely, equity market returns (MKT), short-term reversal factor (ST), and University of Michigan consumer sentiment index (SI), are the predictors most frequently selected by the LASSO and MRS-LASSO models. The results indicate that the financial predictors contain more useful information than the macroeconomic factors for forecasting future stock volatility. Additionally, based on the frequency of selected variables, the MRS-LASSO model can be more powerful than the LASSO model in selecting information that is predictive of future stock volatility.

(3) The expected VRP, constructed by volatility forecasts generated by the LASSO and MRS-LASSO models, can significantly predict future S&P 500 aggregate returns. Moreover, the adjusted R-square value of $\mathrm{VRP}_t^{\mathrm{MRSLASSO}}$ is larger than the value of $\mathrm{VRP}_t^{\mathrm{LASSO}}$ for the short (1- and 2-month-ahead) and long (9- and 12-month-ahead) forecast horizons, implying that the forecasting quality of the MRS-LASSO model is superior to that of the LASSO model from an application perspective.

Our study is related to Fang et al. (2020) and Cheng et al. (2021). The contribution of our research compared to the existing studies on forecasting stock market volatility is threefold. First, this study mainly focuses on stock volatility forecasting using a novel model incorporating regime switching and variable selection. The out-of-sample evaluation methods reveal that the MRS-LASSO model can statistically and economically significantly outperform competing models. Second, we explore the predictability of the MRS-LASSO model and investigate the role of macroeconomic and financial predictors. Our study provides new evidence that financial predictors are more powerful than macroeconomic predictors in forecasting future stock volatility. Third, we construct the expected VRP by using volatility forecasts generated from the LASSO and MRS-LASSO models. Then, we explore the predictability of future MKT based on the expected VRP. This contribution is very significant in terms of the application of this research.

The remainder of the paper is organized as follows. Section 2 presents the methodology, and Section 3 provides information on the data. Section 4 contains the empirical analysis, including the estimation method, forecasting performance, and robustness check. Section 5 discusses the VRP and return predictability, and concluding remarks are provided in Section 6.

2. Methodology

2.1. Realized measures

The superior ex-post variance, RV, is commonly used as the proxy for volatility because it contains less noise (Andersen & Bollerslev, 1997; Barunik et al., 2016; Degiannakis & Filis, 2017; He et al., 2021; Koopman et al., 2005; Liu et al., 2015; Sévi, 2014; Tang et al., 2021; Wang et al., 2016; Xiao et al., 2021). Based on the existing data (e.g., Wang et al., 2018), RV at a monthly frequency is defined as the sum of daily squared returns. For a specific month t, this ex-post measure of variance is given by

$$\mathrm{RV}_t = \sum_{j=1}^{M} r_{t,j}^2, \tag{1}$$

where M is the number of trading days in a specific month, and $r_{t,j}$ represents the jth day return of month t. Additionally, we employ the natural logarithm of RV in the forecasting process for two reasons.[2] First, according to Andersen et al. (2007) and Buncic and Gisler (2017), the RV obtained from Eq. (1) is leptokurtic, and the distribution of logarithmic RV is approximately Gaussian. Second, using the natural logarithm of RV can ensure that the volatility generated from the forecasting model is always positive.

2.2. Autoregressive model with variable selection

We first introduce our benchmark, the AR model, which is widely used in forecasting stock volatility at a monthly frequency and is defined as follows:

$$V_{t+1} = \beta_0 + \sum_{i=0}^{p-1} \beta_{i,1} V_{t-i} + \varepsilon_{t+1}, \tag{2}$$

where V_{t+1} is the logarithmic RV in month t+1, p is the lag order, and ε_{t+1} is the disturbance term. As Wang et al. (2018) and Paye (2012) note, the lag order, p, is equal to 6. This lag length can capture the powerful autocorrelation of stock volatility. Subsequently, we extend the AR model with several macroeconomic and financial predictors to explore their predictability. The AR-X model can be expressed as

$$V_{t+1} = \beta_0 + \sum_{i=0}^{p-1} \beta_{i,1} V_{t-i} + \beta_{i,2} \boldsymbol{X}_{i,t} + \varepsilon_{t+1}, \tag{3}$$

where $\boldsymbol{X}_{i,t}$ represents the ith predictor in month t and $\beta_{i,2}$ denotes the coefficient of the individual predictor. The optimal coefficient, β, is estimated using the maximum

[2] In all subsequent discussions in this study, $V_t = \log(\mathrm{RV}_t)$. We transform the logarithm of forecasts ($\hat{V}_t$) into the original form ($\widehat{\mathrm{RV}}_t$) using the bias-corrected approach of Proietti and Lütkepohl (2013) due to the nonlinear feature of the logarithmic function. Specifically, $\widehat{\mathrm{RV}}_t = \exp(\hat{V}_t + 0.5\sigma_\varepsilon^2)$, where σ^2 refers to the variance of the errors (ε).

likelihood estimation method, and it can be written as[3]

$$\hat{\beta} = \underset{\beta}{\operatorname{argmin}} \frac{1}{T} \sum_{t} \left(V_{t+1} - \beta_0 - \sum_{i=0}^{p-1} \beta_{i,1} V_{t-i} - \beta_{i,2} \boldsymbol{X}_{i,t} \right)^2 . \tag{4}$$

The least absolute shrinkage and selection operator (LASSO), pioneered by Tibshirani (1996), is an excellent method to overcome overfitting issues in forecasting methodology. The LASSO coefficients are estimated by minimizing the log-likelihood function (Eq. (4)) with L_1 parameter penalization. Statistically, the forecasts obtained from LASSO are calculated as

$$\hat{V}_{t+1} = \hat{\beta}_0 + \sum_{i=0}^{p-1} \hat{\beta}_{i,1} V_{t-i} + \hat{\beta}_{i,2} \boldsymbol{X}_{i,t}$$
$$\text{subject to} \sum_{i=1}^{K} |\beta_i| < \psi, \tag{5}$$

or equivalently,

$$\hat{\beta} = \underset{\beta}{\operatorname{argmin}} \left(\frac{1}{T} \sum_{t} \left(V_{t+1} - \beta_0 - \sum_{i=0}^{p-1} \beta_{i,1} \boldsymbol{V}_{t-i} - \sum_{i=1}^{K} \beta_{i,2} \boldsymbol{X}_{i,t} \right)^2 + \lambda \sum_{i=1}^{K} |\beta_i| \right), \tag{6}$$

where K is the number of predictors and ψ (λ) is the tuning parameter that can control the shrinkage estimators in terms of penalty strictness. An increasing value of ψ (λ) represents stricter coefficient selection and thereby causes more coefficient estimates to be zero.

2.3. LASSO with regime-switching technique

To investigate whether the regime-switching technique can improve the accuracy of stock volatility forecasting, we introduce the MRS-LASSO model, which incorporates regime switching and variable selection. Statistically, the MRS-LASSO model is defined as

$$\hat{V}_{t+1,s_t} = \hat{\beta}_{0,s_t} + \sum_{i=1}^{K} \hat{\beta}_{i,s_t} \boldsymbol{X}_{i,t}, \tag{7}$$

where β_{0,s_t} and β_{i,s_t} denote state parameters, S_t represents the unobserved state driven by different conditions, and $S_t = 0$ and $S_t = 1$ are low- and high-volatility regimes, respectively. The regime-switching technique allows volatility persistence to transform or switch into low- and high-volatility regimes. The forecast volatility in month t+1 ($\hat{V}_{t+1}$) can be calculated as the sum of the predictive values multiplied by the respective probability under different regimes, and it is expressed as

$$\hat{V}_{t+1} = P(S_{t+1} = 0|S_t) \times \hat{V}_{t+1,0} + P(S_{t+1} = 1|S_t) \times \hat{V}_{t+1,1}. \tag{8}$$

The transition probability of $P(S_{t+1} = 0|S_t)$ and $P(S_{t+1} = 1|S_t)$ can be evaluated by the Markov chain process. The transition probability matrix of the two-state Markov process is

$$\begin{bmatrix} P^{00} & 1 - P^{00} \\ 1 - P^{11} & P^{11} \end{bmatrix}, \tag{9}$$

where $P^{00} = P(S_t = 0|S_{t-1} = 0)$ and $P^{11} = P(S_t = 1|S_{t-1} = 1)$.

According to Bayes' theorem, we obtain the posterior probability, which can be expressed as

$$P(S_t = 0|V_t) = \frac{P(S_t = 0|S_{t-1}) \times f(V_t|S_t = 0)}{P(S_t = 0|S_{t-1}) \times f(V_t|S_t = 0) + P(S_t = 1|S_{t-1}) \times f(V_t|S_t = 1)}, \tag{10}$$

$$P(S_t = 1|V_t) = 1 - P(S_t = 0|V_t), \tag{11}$$

where $f(V_t|S_t = 0)$ is the maximum likelihood function and is defined as

$$f(V_t|S_t = i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{ -\frac{\left(V_t - \hat{V}_{t,i}\right)^2}{2\sigma^2} \right\}. \tag{12}$$

Considering the MRS-LASSO model as an example, we use the maximum likelihood function with the filtering procedure of Hamilton (1990) and then the smoothing algorithm of Kim (1994) to estimate the coefficient. The estimation procedure of the regime-switching process is shown below.

Step 1: Construct the likelihood function $L(Z|\theta) = P(\theta|Z) = P(\theta|V_{t+1}, \boldsymbol{X})$.

Step 2: Set $P(S(t) = i|S(t)) = 0.5$, $i = 1, 2$; $L_0 = 1$.

Step 3: The fixed transition probability matrix (Eq. (9)) can be iteratively obtained through the following method. Specifically, for $t = 1:T$,

$$P(S(t) = i|S(t-1)) = \sum_{j} P(S(t) = i|S(t-1) = j) P(S(t-1) = j|S(t-1)). \tag{13}$$

$$P(\beta|V_{t+1}, \boldsymbol{X}) = \frac{1}{2\pi\sigma_i} e^{-\frac{1}{2}\left(\frac{V_{t+1} - \beta_{0,s_t} - \sum_{i=1}^{K} \beta_{i,s_t} \boldsymbol{X}_{i,t}}{\sigma_i}\right)^2}. \tag{14}$$

$$P(S(t) = i|S(t)) = \frac{P(S(t) = i|S(t-1)) P\left(V_{t+1}|\beta_{0,s_t} + \sum_{i=1}^{K} \beta_{i,s_t} \boldsymbol{X}_{i,t}, \sigma_i\right)}{\sum_j P(S(t) = j|S(t-1)) P\left(V_{t+1}|\beta_{0,s_t} + \sum_{i=1}^{K} \beta_{i,s_t} \boldsymbol{X}_{i,t}, \sigma_i\right)}. \tag{15}$$

$$P(Z(t)|\theta) = \sum_{i} P(S(t) = i|S(t-1)) P_i(Z(t)|\theta_i). \tag{16}$$

Update $L_t = P(Z(t)|\theta) L_{t-1}$. After iterating, $L_T = L(Z|\theta)$.

Subsequently, we introduce the shrinkage estimator of regression coefficients $\hat{\beta}$ into the regime-switching tech-

[3] Further details can be found in Appendix A.

nique. The MRS-LASSO model extends Eq. (6) with the regime-switching technique, and the specification of $\hat{\beta}_{i,s_t}$ of the MRS-LASSO model is given by

$$\hat{\beta}_{i,s_t} = \underset{\beta}{\operatorname{argmin}} \left(\frac{1}{T} \sum_t \left(V_{t+1} - \beta_0 - \beta_{i,s_t} \boldsymbol{X}_{i,t} \right)^2 + \lambda \sum_{i=1}^{K} \left| \beta_{i,s_t} \right| \right), \tag{17}$$

To obtain the optimal coefficient β_{i,s_t}, we construct the maximum likelihood estimation, $(L(Y|\beta) = P(\beta|Y) = P\left(\beta|V_{t+1}, \boldsymbol{X}_{i,t}\right))$, which can be expressed by

$$P\left(\beta|V_{t+1}, \boldsymbol{X}_{i,t}\right) = \prod_t \frac{1}{\sqrt{2\pi\hat{\sigma}^2}} \exp\left\{ -\frac{\left\| V_{t+1} - \beta_0 - \beta_{i,s_t} \boldsymbol{X}_{i,t} \right\|^2}{2\hat{\sigma}^2} \right\}. \tag{18}$$

The coefficients $\hat{\beta}$ of Eq. (17) can be transformed by[4]

$$\hat{\beta} = \underset{\beta}{\operatorname{argmax}} \left\{ \prod_t \frac{1}{\sqrt{2\pi\hat{\sigma}^2}} \times \exp\left[-\frac{1}{2} \frac{\left(V_{t+1} - \beta_0 - \sum_{i=1}^{K} \beta_i \boldsymbol{X}_{i,t} \right)^2}{\hat{\sigma}^2} \right] \times \left[\prod_i \frac{\lambda}{4\hat{\sigma}^2} \exp\left(-\frac{\lambda}{2\hat{\sigma}^2} |\beta_i| \right) \right]^T \right\}. \tag{19}$$

Two terms form Eq. (19). The first term $\left(\prod_t \frac{1}{\sqrt{2\pi\hat{\sigma}^2}} \exp\left[-\frac{1}{2} \frac{\left(V_{t+1} - \beta_0 - \sum_{i=1}^{K} \beta_i \boldsymbol{X}_{i,t} \right)^2}{\hat{\sigma}^2} \right]\right)$ is equivalent to $P\left(\beta|V_{t+1}, \boldsymbol{X}_{i,t}\right)$, and the second term $\left(\left[\prod_i \frac{\lambda}{4\hat{\sigma}^2} \exp\left(-\frac{\lambda}{2\hat{\sigma}^2} |\beta_i|\right)\right]^T\right)$ can be interpreted by the Laplace prior. The Laplace distribution's probability density function is as follows:

$$f(x|\mu, b) = \frac{1}{2b} \exp\left(-\frac{(x-\mu)}{b}\right), \tag{20}$$

where μ is the location parameter and b is the scale parameter. According to Tibshirani (1996) and Park and Casella (2008), LASSO can be derived by the Bayesian posterior mode under Laplace (e.g., double exponential) priors for coefficient β_i. Let $x = \beta_i$, $\mu = 0$ and $b = \frac{2\hat{\sigma}^2}{\lambda}$. Eq. (20) can be transformed into

$$f\left(\beta_i|0, \frac{2\hat{\sigma}^2}{\lambda}\right) = \frac{\lambda}{4\hat{\sigma}^2} \exp\left(-\frac{\lambda}{2\hat{\sigma}^2} |\beta_i|\right). \tag{21}$$

As a result, the maximum a posteriori estimation (MAE) for MRS-LASSO can be expressed as

$$\text{Posterior}\left(\beta|V_{t+1}, \boldsymbol{X}_{i,t}\right) = \mathrm{P}\left(\beta|V_{t+1}, \boldsymbol{X}_{i,t}\right) \times f\left(\beta_i|0, \frac{2\hat{\sigma}^2}{\lambda}\right), \tag{22}$$

where $\beta_i \sim \text{Laplace}\left(0, \frac{2\hat{\sigma}^2}{\lambda}\right)$ and f represents the Laplace prior for coefficient β. Therefore, the specification of β_{i,s_t} can be written as

$$\hat{\beta}_{i,s_t} = \underset{\beta}{\operatorname{argmax}}\ \text{Posterior}\left(\beta|V_{t+1}, \boldsymbol{X}_{i,t}\right). \tag{23}$$

2.4. Forecast combination

It is difficult for an individual predictor to maintain superior forecasting performance over the long term or under different market conditions due to model uncertainty (Rapach et al., 2010; Zhang et al., 2019b). Therefore, the forecast combination method is introduced and defined as

$$\hat{\mathrm{V}}_{c,t+1} = \sum_{k=1}^{N} \omega_{k,t} \hat{\mathrm{V}}_{k,t+1}, \tag{24}$$

where $\hat{\mathrm{V}}_{c,t+1}$ and $\hat{\mathrm{V}}_{k,t+1}$ represent the forecasts generated by the combination method and AR-type models, respectively, and $\omega_{k,t-1}$ is the combined weight of the kth individual forecast formed in month t.

Inspired by Rapach et al. (2010), five commonly used combination methods are introduced to generate the forecasts: mean combination (mean), median combination (median), trimmed mean combination (trimmed mean), and discount mean squared forecasting error combination (DMSPE (1) and DMSPE (0.9)).[5] Specifically, the combined weights of DMSPE (1) and DMSPE (0.9) can be defined as

$$\omega_{k,t-1} = \frac{\varnothing_{i,t-1}^{-1}}{\sum_{k=1}^{N} \varnothing_{k,t-1}^{-1}}, \tag{25}$$

$$\varnothing_{k,t-1}^{-1} = \sum_{s=m+1}^{t} \xi^{t-s} \left(V_s - \hat{V}_{k,s} \right), \tag{26}$$

where m is the original sample period length and ξ is a discount factor. Following Rapach et al. (2010), we also apply two values of ξ, that is, 1 and 0.9.

Table 1 displays the details of the models investigated in our study.

3. Data description

3.1. Stock data

This study focuses on the S&P 500 index and macroeconomic and financial predictors. The daily closing price of the S&P 500 is collected from the Wind database, covering the period from January 1, 1978, to September 30, 2020, containing 10,872 trading days.[6] Then, we obtain the time series of RV at a monthly frequency covering 513 months after calculating the sum of daily squared returns.

[4] The detailed derivation process can be found in Appendix A.

[5] More technical details regarding the five combination methods can be found in Zhang et al. (2019b).

[6] We choose January 1978 as our starting point for the data because the University of Michigan Consumer Sentiment Index is available beginning in January 1978.

Table 1
Description of forecasting models.

Forecasting model	Model name	Description	Equation
Model 1	AR	Benchmark	Eq. (2)
Model 2	AR-X	Extended with variables	Eq. (3)
Model 3	Mean combination	Combination method	Eq. (24)
Model 4	Median combination	Combination method	Eq. (24)
Model 5	Trimmed mean combination	Combination method	Eq. (24)
Model 6	DMSPE (1) combination	Combination method	Eq. (24)
Model 7	DMSPE (0.9) combination	Combination method	Eq. (24)
Model 8	LASSO	AR with variable selection	Eq. (5)
Model 9	MRS-LASSO	LASSO with regime switching	Eq. (7)

3.2. Macroeconomic and financial predictors

A large number of predictors are included in this research. These predictors are considered in Engle et al. (2013), Conrad and Kleen (2020), Fang et al. (2020), and others, and they include 12 macroeconomic variables: Chicago Fed National Activity Index (CFNAI), corporate profits after tax (CP), CPI, housing starts (HS), industrial production growth rate (IPGR), monetary base (BOG), New Orders Index of the Institute of Supply Management (NO), PPI, real GDP (GDP), real personal consumption (RPC), unemployment rate (UR), and SI. Five financial predictors are also included: default spread (DS), MKT, the news-based implied volatility index (NVIX), ST, and term spread (TS).[7] Conrad and Loch (2015) and Fang et al. (2020) state that real-time data are critical for evaluating forecasting performance. Therefore, we employ the first release data rather than revised data in this research. The 17 macroeconomic and financial predictors are collected from the Federal Reserve Economic Data (FRED, https://fred.stlouisfed.org/), Real-time Data Research Center, CRSP, and the personal webpages of K. R. French and A. Manela. The 17 macroeconomic and financial predictors are available monthly except for GDP and CP.[8] The full sample period runs from January 1978 to September 2020, consisting of 513 months. Table 2 shows the details of all the predictors included in this research. We take the first difference for UR and SI and the log difference for NO and NVIX. Moreover, we consider the CFNAI in levels. The remaining predictors are cleaned by annualized month-over-month percentage changes.

Table 2
Variable description.

Variable	Factor
UR	Calculated as the first difference of the unemployment rate
SI	Calculated as the first difference of the University of Michigan Consumer Sentiment Index
NO	Calculated as the log difference of the New Orders Index of the Institute of Supply Management
CFNAI	Chicago Fed National Activity Index
BOG	Annualized month-over-month percentage changes in the monetary base
PPI	Annualized month-over-month percentage changes in PPI
CPI	Annualized month-over-month percentage changes in CPI
HS	Annualized month-over-month percentage changes in housing starts
IPGR	Annualized month-over-month percentage changes in the industrial production growth rate
CP	Annualized month-over-month percentage changes in corporate profits after tax
GDP	Annualized month-over-month percentage changes in real GDP
RPC	Annualized month-over-month percentage changes in real personal consumption
TS	Term spread
DS	Default spread
MKT	Equity market returns
ST	Short-term reversal factor
NVIX	Calculated as the log difference of the news-based implied volatility index

4. Empirical analysis

4.1. Estimation method

4.1.1. Out-of-sample R^2 test

Out-of-sample predictive quality is vital to market participants and researchers. Therefore, we employ three popular evaluation methods to compare the forecasting quality of the above models. First, the out-of-sample R^2 test of Campbell and Thompson (2008) is frequently applied to evaluate the out-of-sample predictive accuracy and assess the significant differences across different models. The R^2_{OOS} measure is defined as:

$$R^2_{\text{oos}} = 1 - \frac{\text{MSPE}_F}{\text{MSPE}_B}, \tag{27}$$

where MSPE is the mean squared forecasting error, and MSPE_F and MSPE_B represent the volatility from a competing model and benchmark (AR model), respectively. $\text{MSPE}_i = \frac{1}{T-M}\sum_{t=M+1}^{T}(\text{RV}_t - \widehat{\text{RV}}_{t,i})^2$, for $i = F, B$, T, and M are the lengths of the entire sample and the in-sample period, respectively. Furthermore, we introduce the MSPE-adjusted statistic proposed by Clark and West (2007) to identify whether the extended and benchmark models exhibit heterogeneous performance in forecasting stock market volatility. A positive R^2_{OOS} indicates that the forecasts from extended models have lower MSPE than the benchmark model; thus, competing models are superior to the benchmark model.

[7] The economic illustration of 17 factors can be found in Fang et al. (2020) and among others.

[8] As data on GDP and CP are available only quarterly, we transform the quarterly data into monthly data.

4.1.2. Direction-of-change test

The DoC test is also introduced to evaluate out-of-sample performance. According to Degiannakis and Filis (2017), the DoC test plays a fundamental role in timing market participants to allocate assets. This test can determine the fraction of the forecast that has appropriately determined the upward or downward movement of volatility, and it is expressed by

$$P_{t,i} = \begin{cases} 1 \text{ if } \mathrm{RV_t} > \mathrm{RV_{t-1}} \text{ and } \widehat{\mathrm{RV}}_{\mathrm{t-1}} > \mathrm{RV_{t-1}}, \\ 1 \text{ if } \mathrm{RV_t} \leq \mathrm{RV_{t-1}} \text{ and } \widehat{\mathrm{RV}}_{\mathrm{t-1}} \leq \mathrm{RV_{t-1}}, \\ 0 \text{ otherwise}, \end{cases} \tag{28}$$

where $P_{t,i}$ is the indicator value for whether model i correctly determines the movement of volatility. The success ratio (SR) indicates that model i correctly determines the movement of volatility, and it is given by

$$SR_i = \frac{\sum_{t=1}^{N} P_{t,i}}{N} \times 100. \tag{29}$$

Furthermore, we employ the Pesaran and Timmermann (2009) methodology to examine the statistical significance of directional accuracy with the null hypothesis of no directional accuracy.

4.1.3. Economic value analysis

The two methods presented in Sections 4.1.1 and 4.1.2 can be used to statistically compare forecasting accuracy. However, investors, researchers, and decision-makers pay attention to the statistical predictive ability and economic gains of forecasting models. The realized utility framework of Bollerslev et al. (2018), which only relies on volatility forecasts ($\widehat{\mathrm{RV}}$), is used to examine economic gains. Specifically, a mean–variance investor assigns ω_t of his or her assets to the risky asset with r_{t+1} and $(1-\omega_t)$ to the risk-free asset with r_t^f. The expected utility with a constant conditional Sharpe ratio can be written as

$$U(x_t) = W_t\left(x_t SR\sqrt{E_t(\mathrm{RV}_{t+1})} - \frac{\gamma}{2}x_t^2 E_t(\mathrm{RV}_{t+1})\right), \tag{30}$$

where γ denotes the investor's relative risk aversion, $E_t(\mathrm{RV}_{t+1}) = Var(r_{t+1}^e)$, where r_{t+1}^e is the excess return, $r_{t+1} - r_t^f$, and the constant Sharpe ratio, $\mathrm{SR} \equiv E_t\left(r_{t+1}^e\right)/\sqrt{E_t(RV_{t+1})}$. To obtain the optimal portfolio, the investor allocates his or her $x_t^* = E_t\left(r_{t+1}^e\right)/(\gamma E_t(\mathrm{RV}_{t+1}))$ wealth to the risky asset, or alternatively,

$$\omega_t^* = \frac{SR/\gamma}{\sqrt{E_t(\mathrm{RV}_{t+1})}}, \tag{31}$$

where SR/γ denotes the optimal risk target. Hence, the expected utility regarding the optimal target portfolio can be defined as

$$U\left(\omega_t^*\right) = \frac{SR^2}{2\gamma}W_t = \frac{1}{2} \times SR \times \frac{SR}{\gamma}W_t, \tag{32}$$

where 1/2 refers to half of the expected return not lost to the disutility of risk, and $SR \times \frac{SR}{\gamma}$ is the expected excess return. However, the investor cannot observe $E_t(\mathrm{RV}_{t+1})$ in practice. As a result, the volatility forecast $(\widehat{\mathrm{RV}}_{t+1})$ replaces $E_t(\mathrm{RV}_{t+1})$, resulting in the average expected utility of

$$\overline{U}\left(\widehat{\mathrm{RV}}_{t+1}\right) = \frac{1}{q}\sum_{t=m+1}^{m+q-1}\frac{SR^2}{\gamma}\left(\frac{\sqrt{\mathrm{RV}_{t+1}}}{\sqrt{\widehat{\mathrm{RV}}_{t+1}}} - \frac{1}{2}\frac{\mathrm{RV}_{t+1}}{\widehat{\mathrm{RV}}_{t+1}}\right). \tag{33}$$

Following Bollerslev et al. (2018) and Liang et al. (2020), we set the annualized Sharpe ratio and the coefficient of relative risk aversion as $SR = 0.4$ and $\gamma = 2$, respectively. Consequently, $U\left(x_t^*\right) = 4\%W_t$, implying that the investor would be willing to pay 4% of his or her wealth to gain access to the ω_t^* portfolio of the risky asset rather than simply investing in the risk-free asset. Parallel to Eq. (33), a risk model that perfectly predicts the RVs delivers a realized utility of 8%-4%=4%.

4.2. Forecasting performance

4.2.1. Which model can outperform competing models?

Our database ranges from January 1978 to September 2020 (513 months). We employ the first 10% of observations as the estimation period from January 1978 to March 1982 (51 months) and the last 90% of observations as the out-of-sample forecasting sample from April 1982 to September 2020 (462 months).[9] The out-of-sample volatility forecasts are generated by the expanding (recursive) window method.

We compare the predictive volatility with the ex-post realized variance and consider three prevailing evaluation methods to assess the forecasting performance. The results of the one-month-ahead out-of-sample evaluation tests are reported in Table 3. Several interesting findings are highlighted.

First, the out-of-sample R^2 results show that LASSO and MRS-LASSO can yield significantly larger R^2 values than the competing models. Moreover, individual predictors and the combined method cannot successfully predict future stock volatility, except for NVIX. One possible reason is that a single variable cannot maintain predictability during the long term (462 months) or across changing market conditions. Second, the SR values of each forecasting model are over 60%, implying that each model has powerful forecasting ability for stock volatility. Additionally, the SR values of LASSO and MRS-LASSO are 70.4% and 71.9%, respectively, providing empirical evidence that the MRS-LASSO model outperforms the competing models. Third, the economic analysis results show that the forecasting model with regime switching and variable selection can yield a 3.449% utility. This result suggests that the model can produce superior economic value in an actual portfolio exercise.

4.2.2. Market conditions and forecasting performance

Furthermore, we further depict the cumulative difference (called CumSFE and evaluated as $\mathrm{CumSFE} = \sum_{t=M+1}^{T}\left(\left(\widehat{\mathrm{RV}}_t^{\mathrm{model}} - \mathrm{RV}_t\right)^2 - \left(\widehat{\mathrm{RV}}_t^{\mathrm{bench}} - \mathrm{RV}_t\right)^2\right)$) of LASSO and MRS-LASSO to explore the forecasting performance under different market conditions. If the value of CumSFE

[9] We consider alternative out-of-sample periods (410 and 436 months), and the results are similar.

Table 3
Out-of-sample performance.

Regressions	Out-of-sample R^2			DoC test			Economic analysis
	R^2_{OOS}(%)	Adj. MSPE	*P*-value	SR (%)	Stat. value	*P*-value	
Bench	–	–	–	64.300	6.773	0.000	2.751
AR-UR	−0.807	−0.469	0.680	64.900	7.027	0.000	2.798
AR-SI	0.196	0.696	0.243	64.900	7.118	0.000	2.739
AR-NO	−0.180	−0.987	0.838	64.700	7.002	0.000	2.728
AR-CFNAI	−0.411	−0.145	0.558	64.100	6.688	0.000	2.766
AR-BOG	−2.200	−1.278	0.899	64.100	6.659	0.000	2.749
AR-PPI	−0.237	−0.756	0.775	63.800	6.603	0.000	2.759
AR-CPI	−0.139	−0.641	0.739	63.600	6.433	0.000	2.728
AR-HS	0.024	0.413	0.340	64.300	6.773	0.000	2.747
AR-IPGR	−1.445	−0.952	0.829	63.800	6.603	0.000	2.771
AR-CP	−0.357	−0.364	0.642	64.900	6.998	0.000	2.690
AR-GDP	−0.018	−0.037	0.515	64.100	6.659	0.000	2.732
AR-RPC	0.126	0.971	0.166	64.100	6.688	0.000	2.757
AR-TS	−0.074	−0.302	0.619	64.100	6.659	0.000	2.758
AR-DS	−0.585	−0.852	0.803	65.600	7.224	0.000	2.690
AR-MKT	0.242	1.292	0.098	65.400	7.318	0.000	2.760
AR-ST	−0.458	−0.622	0.733	64.300	6.744	0.000	2.717
AR-NVIX	0.804	1.595	0.055	64.500	6.887	0.000	2.751
MEAN	−0.091	−0.426	0.665	64.300	6.773	0.000	2.750
MEDIAN	−0.011	−0.176	0.570	64.300	6.773	0.000	2.750
TMC	−0.060	−0.399	0.655	64.300	6.773	0.000	2.749
DMSPE1	−0.063	−0.299	0.618	64.300	6.773	0.000	2.750
DMSPE2	−0.059	−0.275	0.608	64.300	6.773	0.000	2.750
LASSO	22.929	2.267	0.012	70.400	8.805	0.000	3.330
MRS-LASSO	43.617	1.840	0.033	71.900	9.491	0.000	3.449

Notes: The table presents the evaluation of the out-of-sample performance of the models based on the out-of-sample R^2 test, DoC test, and economic value analysis. The first out-of-sample forecast begins in April 1982, and the forecasting window covers 462 months.

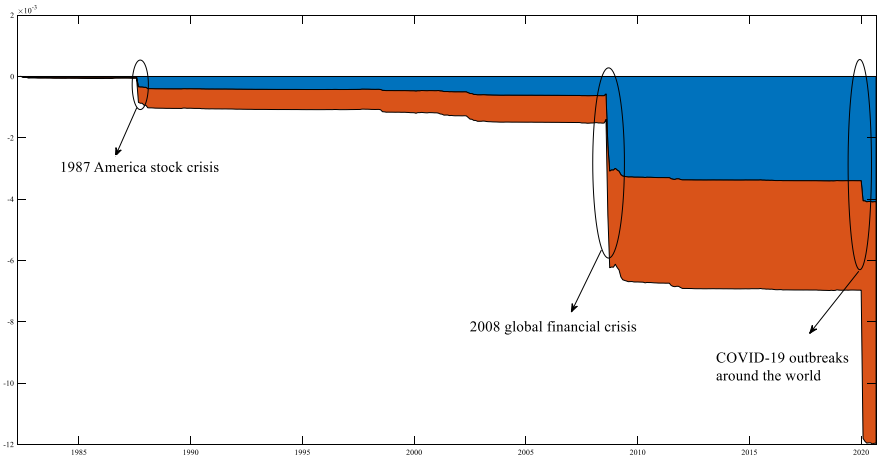

Fig. 2. The CumSFE of the LASSO and MRS-LASSO models.

is negative, the forecasting model is superior to the benchmark model. Fig. 2 shows the values of CumSFE. The CumSFE of the MRS-LASSO model is always smaller than that of LASSO, implying that the MRS-LASSO model can outperform LASSO in forecasting future stock volatility. Additionally, CumSFE declined sharply during three incidents: the American stock crisis in 1987, the global financial crisis in 2008, and the financial turmoil caused by the COVID-19 pandemic in 2020. The results provide evidence that the forecasting performance of the MRS-LASSO model is always powerful during a crisis.

4.2.3. Business cycle and forecasting ability

According to Paye (2012) and Choudhry et al. (2016), among others, the business cycle plays a vital role in stock volatility. Therefore, we divide the out-of-sample volatility forecasts into recession and expansion periods by using NBER business cycle data. The out-of-sample R^2 results are reported in Table 4. The forecasting abilities of the LASSO and MRS-LASSO models are roughly equivalent during a recession. In contrast, the MRS-LASSO model can significantly outperform other models during an expansion. The results indicate that the forecasting ability of MRS-LASSO is more powerful during expansion periods when the economy starts to prosper. There

Table 4
Out-of-sample performance during recession and expansion periods.

Regressions	Recession			Expansion		
	$R^2_{OOS}(\%)$	Adj. MSPE	*p*-value	$R^2_{OOS}(\%)$	Adj. MSPE	*p*-value
MEAN	−226.265	1.295	0.098	−0.010	−0.094	0.537
MEDIAN	0.107	0.555	0.289	−0.003	−0.162	0.564
TMC	0.074	0.357	0.360	−0.020	−0.252	0.599
DMSPE1	−239.541	1.309	0.095	−0.010	−0.085	0.534
DMSPE2	−220.759	1.297	0.097	−0.006	−0.045	0.518
LASSO	56.187	1.825	0.034	10.364	1.272	0.102
MRS-LASSO	55.875	1.819	0.034	39.171	2.344	0.010

Notes: The table represents the evaluation of the out-of-sample performance of models based on the out-of-sample R^2 test during recession and expansion periods.

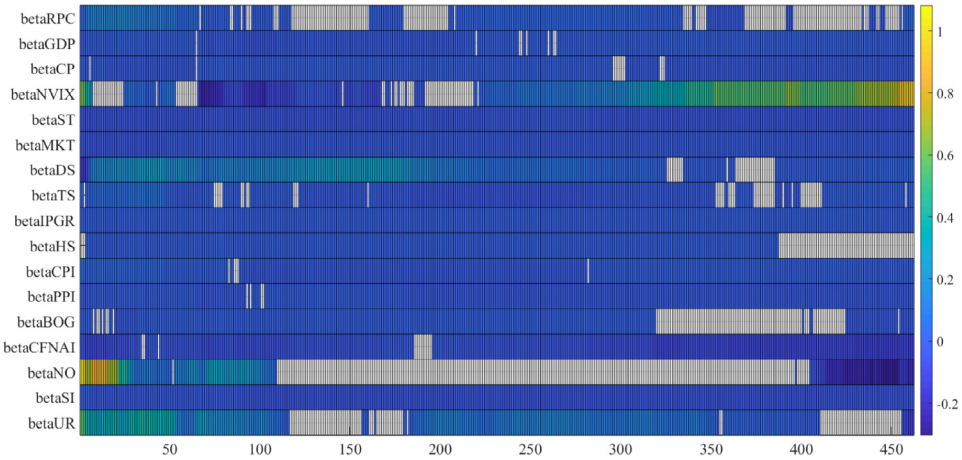

Fig. 3. Variable selection of the LASSO model.

are two possible and reliable reasons for such promising results. First, the difference in forecasting performance can be attributed to the length of the expansion and recession periods. We generate 462 monthly volatility forecasts that include 49 and 413 months during recession and expansion periods, respectively. Second, the regime-switching technique can better capture the volatility dynamic during an expansion period since market participants would avoid higher market risk and require higher risk premiums when the economy is in recession.

Furthermore, we find that the adjusted MSPE (Adj. MSPE) values of the LASSO and MRS-LASSO models are associated with low *p*-values.[10] We have the following possible explanations. First, the Adj.MSPE statistic proposed by Clark and West (2007) is defined as $f_t = (\mathrm{RV}_t - \widehat{\mathrm{RV}}_{t,\mathrm{bench}})^2 - \left(\mathrm{RV}_t - \widehat{\mathrm{RV}}_{t,\mathrm{model}}\right)^2 + \left(\widehat{\mathrm{RV}}_{t,\mathrm{bench}} - \widehat{\mathrm{RV}}_{t,\mathrm{model}}\right)^2$. Low Adj. MSPE statistics can lead to *p*-values that are not significant. Second, we generate 462 monthly volatility forecasts, including 49 and 413 months for recession and expansion periods, respectively. The length of forecasts over the recession and expansion periods may well cause low *p*-values. Finally, the values of R^2_{OOS}, generated from the MRS-LASSO model, are always significant in both recession and expansion periods.

[10] We are grateful to an anonymous referee for this constructive suggestion.

4.2.4. The role of variable selection

The shrinkage method can be used to efficiently select variables and improve accuracy. Hence, we further investigate which and how many variables are selected by the LASSO and MRS-LASSO models from April 1982 to September 2020 (out-of-sample period). Specifically, the coefficients of some variables can be set to zero by the shrinkage method. In other words, the LASSO and MRS-LASSO models select some variables with nonzero coefficients. Figs. 3–5 show the variable selection results based on the LASSO and MRS-LASSO models. Table 5 represents the frequency of selected variables used in the forecasting process over the out-of-sample period.

Based on these figures and tables, several significant findings are summarized below. First, we observe that MKT, ST, and SI are the factors most frequently selected by LASSO. Moreover, MRS-LASSO selects BOG, HS, MKT, SI, IPGR, and ST. These findings indicate that (1) MKT, ST, and SI are the predictors most frequently selected by LASSO and MRS-LASSO; (2) financial variables are more efficient than macroeconomic factors for forecasting future stock volatility; and (3) based on the frequency of selected factors, the MRS-LASSO model can more powerfully select predictive information for future stock volatility than the LASSO model. For example, NO is used in approximately 36% of the LASSO model's monthly prediction models. In contrast, the MRS-LASSO model selects NO at frequencies

Table 5
Variable selection frequencies.

Factors	LASSO		MRS-LASSO		
	Variables	Frequency	Variables	Frequency 0	Frequency 1
Macroeconomic factors	UR	0.771	UR	0.539	0.946
	SI	**0.994**	SI	0.820	**1.000**
	NO	0.361	NO	0.519	0.652
	CFNAI	0.972	CFNAI	0.764	0.959
	BOG	0.762	BOG	**0.937**	0.773
	PPI	0.991	PPI	0.864	0.983
	CPI	0.989	CPI	0.812	0.972
	HS	0.831	HS	**0.998**	0.874
	IPGR	0.913	IPGR	0.879	**1.000**
	CP	0.974	CP	0.887	0.991
	GDP	0.983	GDP	0.877	0.996
	RPC	0.643	RPC	0.812	0.887
Financial factors	TS	0.892	TS	0.803	0.968
	DS	0.931	DS	0.788	0.918
	MKT	**1.000**	MKT	**0.913**	**1.000**
	ST	**1.000**	ST	0.820	**1.000**
	NVIX	0.846	NVIX	0.799	0.937

Notes: This table reports the variable selection frequency of all predictors in each one-month-ahead forecasting step by LASSO and MRS-LASSO across the out-of-sample period (covering 462 months). Frequency 0 (1) represents the variable selection frequency of state 0 (1) from the MRS-LASSO model. Bold and underlined text indicates the most frequently selected factors.

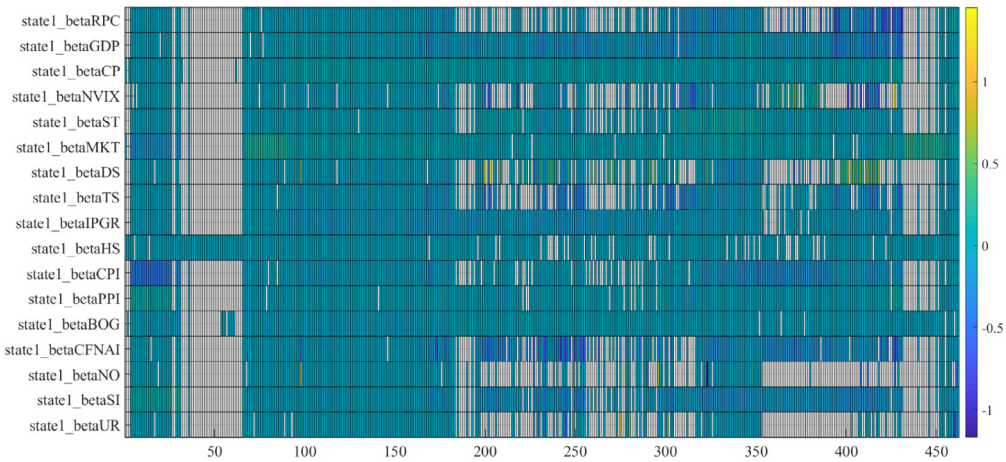

Fig. 4. The variable selection of State 1 based on MRS-LASSO.

of 51% and 65% under high- and low-volatility regimes, respectively.

Second, we observe that the frequencies of selected predictors, except for HS and BOG, for each variable are higher during the high-volatility regime than during the low-volatility regime. These results are consistent with Fig. 2 and reconfirm the superior of the MRS-LASSO model that may be reflected in the high-volatility regime.

Third, Figs. 3–5 depict the estimated coefficients of the LASSO and MRS-LASSO models from April 1982 to September 2020 (462 months, out-of-sample period). In these figures, the color of each rectangle reflects the coefficient estimates for the predictors that we consider. Moreover, each rectangle represents the one-month-ahead volatility forecasts. None of the individual predictors that we consider (except MKT and ST) maintain predictability over the out-of-sample period. This result is consistent with the work of Rapach et al. (2010) and Zhang et al. (2019b). We can also conclude that the MRS-LASSO model can extract more predictive information for future stock volatility than the LASSO model. Additionally, the predictive power of the MRS-LASSO model is reflected in the results for the high-volatility regime.

4.3. Robustness check

4.3.1. Alternative evaluation method

In this subsection, we consider another evaluation method, the MCS test of Hansen et al. (2011), which is widely used in the volatility forecasting literature (Barunik et al., 2016). We set the confidence level, α, at 0.25 to

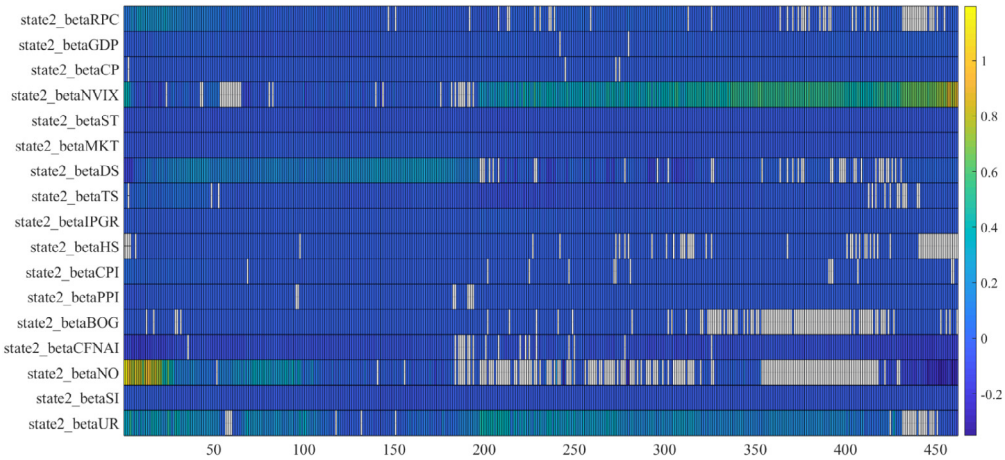

Fig. 5. The variable selection of State 2 based on MRS-LASSO.

Table 6
Out-of-sample performance with an alternative evaluation method.

Regressions	QLIKE		MSE		MAE		HMSE		HMAE	
	Range	SeimQ	Range	SeimQ	Range	SeimQ	Range	SeimQ	Range	SeimQ
Bench	0.099	0.038	**0.266**	0.096	0.001	0.004	0.000	0.001	0.000	0.000
MEAN	0.099	0.038	**0.266**	0.076	0.001	0.004	0.000	0.001	0.000	0.000
MEDIAN	0.099	0.038	**0.266**	0.096	0.001	0.004	0.000	0.001	0.000	0.000
TMC	0.099	0.038	**0.266**	0.096	0.001	0.004	0.000	0.001	0.000	0.000
DMSPE1	0.099	0.038	**0.266**	0.076	0.001	0.004	0.001	0.001	0.000	0.000
DMSPE2	0.099	0.038	**0.265**	0.076	0.001	0.004	0.000	0.001	0.000	0.000
LASSO	0.099	0.038	**0.266**	0.192	0.045	0.045	**0.801**	**0.801**	0.195	0.195
MRS-LASSO	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**

Notes: This table reports the out-of-sample performance based on the MCS test. The p-values are calculated according to the range and semiquadratic (SeimQ) statistics. Bold indicates p-values > 0.25, and a p-value of 1 is indicated in bold and underlined. The rolling window method is employed for forecasting; in this table, the forecast window covers 462 months.

obtain the best model set. Moreover, the stationary bootstrap approach[11] is used to evaluate the interpretation of the MCS test p-value. This implies that the predictive models have better out-of-sample prediction performance when their MCS test p-values are over 0.25. To assess this performance, we introduce five loss functions expressed by

$$\text{QLIKE} = q^{-1} \sum_{t=m+1}^{m+q} (\ln\left(\widehat{\text{RV}_t}\right) + (\frac{\text{RV}_t}{\widehat{\text{RV}_t}})), \tag{34}$$

$$\text{MSE} = q^{-1} \sum_{t=m+1}^{m+q} (\text{RV}_t - \widehat{\text{RV}_t})^2, \tag{35}$$

$$\text{MAE} = q^{-1} \sum_{t=m+1}^{m+q} \left|\text{RV}_t - \widehat{\text{RV}_t}\right|, \tag{36}$$

$$\text{HMSE} = q^{-1} \sum_{t=m+1}^{m+q} \left(1 - \widehat{\text{RV}_t}/\text{RV}_t\right)^2, \tag{37}$$

$$\text{HMAE} = q^{-1} \sum_{t=m+1}^{m+q} \left|1 - \widehat{\text{RV}_t}/\text{RV}_t\right|, \tag{38}$$

where m and q represent the length of the in- and out-of-sample evaluation periods, respectively.

Table 6 reports the results of the MCS test p-values for each model. We observe that the MRS-LASSO model can always yield the largest MCS p-value of 1. Therefore, our proposed model incorporating regime switching and variable selection can more successfully predict future stock volatility than the competing models.

4.3.2. Subsample analysis

To further confirm that our results are convincing over time, the entire sample is divided into two equal subsample lengths around March 1999. One subsample runs from January 1978 to March 1999, while the other ranges from April 1999 to September 2020. The forecasting window covers 205 months for subsamples I and II. Table 7 represents the results of the out-of-sample performance for each model in the subsample analysis. MRS-LASSO can generate higher R^2_{OOS}, SR, and economic gains than the competing models. The results reconfirm the previous findings that the MRS-LASSO model outperforms other models at predicting future equity market volatility.

[11] Further technical details regarding the evaluation of the MCS test p-value can be found in Hansen et al. (2011).

Table 7
Out-of-sample performance with subsample analysis.

Regressions	Out-of-sample R^2			DoC test			Economic analysis
	R^2_{OOS}(%)	Adj. MSPE	*p*-value	SR (%)	stat. value	*p*-value	
Subsample I: January 1978 to March 1999							
Bench	–	–	–	63.100	3.962	0.000	3.012
MEAN	−0.124	−0.806	0.790	63.100	3.962	0.000	2.993
MEDIAN	−0.029	−0.845	0.801	63.100	3.962	0.000	3.008
TMC	−0.119	−0.879	0.810	63.100	3.962	0.000	2.995
DMSPE1	−0.127	−0.807	0.790	63.100	3.962	0.000	2.993
DMSPE2	−0.119	−0.784	0.784	63.100	3.962	0.000	2.994
LASSO	6.789	1.880	0.030	68.000	5.126	0.000	3.399
MRS-LASSO	10.353	1.894	0.029	68.500	5.275	0.000	3.514
Subsample II: April 1999 to September 2020							
Bench	–	–	–	61.600	3.930	0.000	2.335
MEAN	−2.148	−1.012	0.844	61.100	3.805	0.000	2.374
MEDIAN	−0.507	−0.966	0.833	61.600	3.930	0.000	2.343
TMC	−1.500	−1.177	0.880	61.100	3.805	0.000	2.358
DMSPE1	−1.909	−0.922	0.822	61.100	3.805	0.000	2.375
DMSPE2	−1.886	−0.902	0.817	61.100	3.805	0.000	2.376
LASSO	47.069	2.019	0.022	76.400	7.605	0.000	3.429
MRS-LASSO	48.327	2.042	0.021	76.400	7.605	0.000	3.438

Notes: The table represents the evaluation of the out-of-sample performance of models based on the out-of-sample R^2 test, DoC test, and economic value analysis for subsample analysis.

4.3.3. Alternative forecasting horizon

Investors concentrate not only on short-term forecast performance but also on medium-term and long-term predictive abilities. Thus, this paper further investigates whether the MRS-LASSO model consistently exhibits superior out-of-sample performance for 3-month, 6-month, 9-month, and 12-month forecasting horizons. The empirical results for different medium-term and long-term forecasting horizons are shown in Table 8. The robust results show that the value of R^2_{oos} and economic gains obtained from the MRS-LASSO model are higher than those obtained from the competing models for medium-term and long-term horizons (i.e., h=3, 6, 9, and 12). Hence, the MRS-LASSO model can successfully predict future stock volatility.

4.3.4. Crude oil futures market

We re-estimate the out-of-sample forecasting ability of the MRS-LASSO model by considering an alternative financial market, West Texas Intermediate (WTI), a widely used proxy for crude oil futures. The daily prices of front-month WTI crude oil futures are collected from the Energy Information Administration (EIA, https://www.eia.gov/) to evaluate the monthly RV from April 1983 to September 2020, with 450 months in total. We generate out-of-sample volatility forecasts via the recursive window method, and the first predictive volatility starts in November 1987.

The empirical results for out-of-sample performance are shown in Table 9. The MRS-LASSO model can yield higher R^2_{oos} and SR than the competing models, implying that the MRS-LASSO model incorporating regime switching and variable selection can improve the accuracy of volatility forecasting. Moreover, the MRS-LASSO model can generate higher economic gains. Hence, the MRS-LASSO model can improve volatility forecasting accuracy for stock and crude oil futures markets.

4.3.5. Rolling window method

The rolling window method is another popular approach to generate volatility forecasts (Liang et al., 2020; Ma et al., 2017; Paye, 2012; Sévi, 2014; Wang et al., 2020, 2018). Specifically, we set the first 10% of the observations as the estimation period (51 months) and the last 462 months as the out-of-sample period. The forecasting performance of the rolling window method can be found in Table 10. The LASSO and MRS-LASSO models can generate the largest R^2_{OOS} (%), SR (%), value, and realized utility. The results confirm that our findings are robust to the rolling window method.

5. VRP estimation and return predictability

According to Bollerslev et al. (2009) and Drechsler and Yaron (2011), among others, the VRP is closely tied to economic uncertainty and is defined as the difference between squared implied volatility and the conditional variance of the equity market. Following Conrad and Loch (2015), the VRP at a monthly frequency can be expressed as

$$\mathrm{VRP}_t = \mathrm{IV}_t^2 - E\left(\mathrm{RV}_{t+1}\right), \tag{39}$$

where IV is the risk-neutral expected variance in month t and $E(\mathrm{RV}_{t+1})$ represents the physical measure of the expected realized variance. In our study, we measure the IV by the end-of-month value of the squared VIX index, which is proposed by the Chicago Board of Options Exchange[12] (CBOE). Moreover, we employ RV_t to assess $E(\mathrm{RV}_{t+1})$ since the realized measure follows a martingale difference sequence (Bollerslev et al., 2009; Conrad & Loch, 2015).[13] Additionally, Bekaert and Hoerova (2014)

[12] http://www.cboe.com/vix.

[13] Note that in this section, we employ the original RV to construct the VRP rather than the logarithmic RV.

Table 8
Out-of-sample performance with different horizons.

Regressions	Out-of-sample R^2			Economic analysis
	R^2_{OOS}(%)	Adj. MSPE	p-value	
h = 3				
Bench	–	–	–	3.250
MEAN	0.007	0.288	0.387	3.245
MEDIAN	−0.029	−0.294	0.615	3.251
TMC	−0.054	−0.037	0.515	3.253
DMSPE1	−0.062	0.007	0.497	3.245
DMSPE2	−0.042	0.064	0.474	3.245
LASSO	26.353	2.434	0.007	3.462
MRS-LASSO	66.699	2.580	0.005	3.730
h = 6				
Bench	–	–	–	3.511
MEAN	−0.684	−0.755	0.775	3.516
MEDIAN	−0.278	−0.943	0.827	3.513
TMC	−0.566	−0.890	0.813	3.516
DMSPE1	−0.569	−0.616	0.731	3.517
DMSPE2	−0.496	−0.529	0.701	3.518
LASSO	18.646	2.208	0.014	3.588
MRS-LASSO	57.051	3.740	0.000	3.844
h = 9				
Bench	–	–	–	3.629
MEAN	−3.883	−0.991	0.839	3.634
MEDIAN	−0.363	−0.866	0.807	3.630
TMC	−0.522	−0.664	0.747	3.633
DMSPE1	−3.816	−0.940	0.826	3.634
DMSPE2	−3.645	−0.819	0.794	3.636
LASSO	9.598	2.471	0.007	3.681
MRS-LASSO	12.638	2.937	0.002	3.715
h = 12				
Bench	–	–	–	3.666
MEAN	−0.896	−0.549	0.709	3.671
MEDIAN	−0.263	−0.571	0.716	3.667
TMC	−0.129	−0.106	0.542	3.670
DMSPE1	−0.813	−0.447	0.673	3.672
DMSPE2	−0.562	−0.228	0.590	3.673
LASSO	9.275	2.630	0.004	3.722
MRS-LASSO	71.270	6.206	0.000	3.895

Notes: The table represents the evaluation of the out-of-sample performance of models based on the out-of-sample R^2 test and economic value analysis for different forecasting horizons.

Table 9
Out-of-sample performance for the crude oil futures market.

Regressions	Out-of-sample R^2			DoC test			Economic analysis
	R^2_{OOS}(%)	Adj. MSPE	p-value	SR (%)	Stat. value	p-value	
Bench	–	–	–	60.100	5.044	0.000	2.960
MEAN	−0.361	−0.333	0.631	60.100	5.130	0.000	2.984
MEDIAN	−0.095	−0.580	0.719	60.100	5.086	0.000	2.965
TMC	−0.345	−0.769	0.779	60.100	5.130	0.000	2.976
DMSPE1	−0.330	−0.193	0.577	60.100	5.130	0.000	2.986
DMSPE2	−0.358	−0.268	0.606	60.100	5.130	0.000	2.984
LASSO	63.845	1.282	0.100	67.000	6.702	0.000	3.526
MRS-LASSO	66.441	1.293	0.098	67.300	6.808	0.000	3.552

Notes: The table represents the evaluation of the out-of-sample performance of models based on the out-of-sample R^2 test, DoC test, and economic value analysis for the crude oil market.

state that replacing $E(\mathrm{RV}_{t+1})$ with RV_t is not appropriate. Therefore, we construct the expected VRP with conditional variance generated using the forecasts obtained from the LASSO and MRS-LASSO models. The VRP can be defined as

$$\mathrm{VRP}_t^{\mathrm{LASSO}} = \mathrm{IV}_t^2 - \widehat{\mathrm{RV}}_{t+1}^{\mathrm{LASSO}}, \tag{40}$$

$$\mathrm{VRP}_t^{\mathrm{MRSLASSO}} = \mathrm{IV}_t^2 - \widehat{\mathrm{RV}}_{t+1}^{\mathrm{MRSLASSO}}, \tag{41}$$

where $\widehat{\mathrm{RV}}_{t+1}^{\mathrm{LASSO}}$ and $\widehat{\mathrm{RV}}_{t+1}^{\mathrm{MRSLASSO}}$ denote the forecasts of RVs generated from the LASSO model and MRS-LASSO model, respectively, from January 1991 to September 2020, including 369 observations at the monthly frequency since the VIX is available at the same time.

Table 10
Out-of-sample performance of the rolling window method.

Regressions	Out-of-sample R^2			DoC test			Economic analysis
	R^2_{OOS}(%)	Adj. MSPE	p-value	SR (%)	stat. value	p-value	
Bench	–	–	–	66.522	7.298	0.000	2.934
MEAN	0.015	0.925	0.178	67.609	7.763	0.000	2.960
MEDIAN	0.032	0.788	0.215	66.739	7.369	0.000	2.932
TMC	0.224	1.057	0.145	67.174	7.543	0.000	2.946
DMSPE1	0.031	0.946	0.172	67.609	7.763	0.000	2.960
DMSPE2	0.043	0.974	0.165	67.391	7.660	0.000	2.961
LASSO	52.562	1.655	0.049	74.348	10.470	0.000	3.546
MRS-LASSO	56.330	1.728	0.042	76.087	11.217	0.000	3.662

Notes: The table represents the evaluation of the out-of-sample performance of models based on the out-of-sample R^2 test, DoC test, and economic value analysis for the rolling window method.

Table 11
Return predictability regressions.

Variance premium	Horizon	1	2	3	6	9	12
	Constant	−2.269	−0.444	1.015	2.260	3.400	4.074
		(−0.850)	(−0.220)	(0.580)	(1.800)	(3.530)	(4.610)
VRP	VRP	0.710	0.528	0.383	0.256	0.312	0.086
		(5.980)	(5.570)	(4.400)	(4.620)	(3.890)	(2.740)
	Adj.R^2	23.690	25.460	19.950	16.490	6.350	3.460
	Constant	−8.335	−4.777	−1.986	0.011	2.275	3.256
		(−2.950)	(−2.380)	(−1.110)	(0.010)	(2.270)	(3.500)
VRP^{LASSO}	VRP	0.940	0.686	0.487	0.343	0.179	0.119
		(8.460)	(7.430)	(5.390)	(6.300)	(5.170)	(3.550)
	Adj.R^2	27.470	28.450	**21.400**	**19.640**	7.800	4.470
	Constant	−10.090	−5.777	−2.392	−0.429	1.876	2.959
$VRP^{MRSLASSO}$		(−3.840)	(−2.860)	(−1.290)	(−0.330)	(1.890)	(3.290)
	VRP	1.057	0.751	0.512	0.372	0.206	0.140
		(9.130)	(6.860)	(4.660)	(6.440)	(5.840)	(4.420)
	Adj.R^2	**28.990**	**28.480**	19.710	19.200	**8.600**	**5.090**

Notes: This table shows the coefficient estimation for different horizons. The sample period from January 1991 to September 2020 includes 369 observations at a monthly frequency. The bold number is the highest Adj. R^2 value for each horizon.

Subsequently, we explore the predictability of the expected VRP for future MKT using a simple linear model expressed as

$$\frac{1}{h}\sum_{j=1}^{h} r_{t+j}^{ex} = a_h + b_h Z_t + u_{t,t+h}, \tag{42}$$

where $Z_t \in \left\{\mathrm{VRP}_t, \mathrm{VRP}_t^{\mathrm{LASSO}}, \mathrm{VRP}_t^{\mathrm{MRSLASSO}}\right\}$ and r_{t+j}^{ex} are the excess returns and are evaluated as the difference in the S&P 500 return and the 1-month T-bill rate. The coefficients of a_h and b_h are estimated via the OLS method with Newey–West robust standard errors. The results of the coefficient estimation for different horizons h are reported in Table 11.

We find that the coefficient estimates of the expected VRP ($\mathrm{VRP}_t, \mathrm{VRP}_t^{\mathrm{LASSO}}, \mathrm{VRP}_t^{\mathrm{MRSLASSO}}$) are significantly positive for various horizons, implying that VRP has powerful predictive value for future stock returns. Furthermore, the adjusted-R^2 values of the expected VRP based on the predictive volatility from the LASSO and MRS-LASSO models are higher than the initial VRP constructed by the actual volatility for each forecasting horizon. This indicates that the LASSO and MRS-LASSO models can successfully predict future stock volatility from the perspectives of VRP and return predictability. The adjusted R-squared values of $\mathrm{VRP}_t^{\mathrm{MRSLASSO}}$ are larger than those of $\mathrm{VRP}_t^{\mathrm{LASSO}}$ for the short and long forecast horizons (1-, 2-, 9-, 12-). This result suggests that the expected VRP generated from the MRS-LASSO model can significantly better predict the aggerated equity future return than the previous measures.

6. Conclusion

This paper focuses on developing a novel forecasting model, MRS-LASSO, that incorporates a regime-switching technique and variable selection to predict future stock volatility. Following Conrad and Kleen (2020), Engle et al. (2013), Fang et al. (2020), and Paye (2012), 12 macroeconomic and 5 financial predictors are used to explore the predictability of each forecasting model. The out-of-sample evaluation results indicate that the MRS-LASSO model can successfully better predict future stock volatility than the competing models, including the AR model, the AR model with individual predictors, and the LASSO model. Furthermore, financial factors are more powerful than macroeconomic predictors in forecasting future stock volatility. The LASSO and MRS-LASSO models most frequently select three variables MKT, ST, and SI.

We introduce the expected VRP to explore the return predictability of the volatility forecasts generated from the LASSO and MRS-LASSO models as an application of volatility forecasts. The results show that $\mathrm{VRP}_t^{\mathrm{MRSLASSO}}$ is

more powerful than $\mathrm{VRP}_t^{\mathrm{LASSO}}$ and VRP_t when predicting future stock returns in the short horizon and long horizon (1, 2, 9, and 12 months ahead). Overall, the MRS-LASSO model can consistently predict future stock volatility as validated by statistical results and application.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Assume a simple linear regression, $Y_t = \beta X_t + \varepsilon_t$, $t = 1, \ldots, T$. Construct the maximum likelihood, $L\left(\beta, \sigma^2 | Y, X\right) = f\left(Y, X | \beta, \sigma^2\right)$, and the optimal estimation of β can be defined as

$$\underset{\beta}{\operatorname{argmax}} L\left(\beta, \sigma^2 | Y, X\right) = \underset{\beta}{\operatorname{argmin}} E\left[(Y_t - \beta X_t)^2\right], \tag{A.1}$$

Then,

$$\hat{\beta} = \underset{\beta}{\operatorname{argmin}} E\left[(Y_t - \beta X_t)^2\right] = \underset{\beta}{\operatorname{argmin}} \frac{1}{T} \sum_t (Y_t - \beta X_t)^2, \tag{A.2}$$

where $E\left[(Y_t - \beta X_t)^2\right]$ represents the optimal estimation of σ^2 ($\varepsilon_t \sim N(0, \sigma^2)$).

LASSO extends Eq. (A.2) with a penalty function, which is defined as

$$\hat{\beta} = \underset{\beta}{\operatorname{argmin}} \left(\frac{1}{T} \sum_t (Y_t - \beta X_t)^2 + \lambda \sum_{i=}^{K} |\beta_i| \right), \tag{A.3}$$

where λ is the nonnegative regularization parameter serving as the penalty function of β. Eq. (A.3) can be written as

$$\hat{\beta} = \underset{\beta}{\operatorname{argmax}} \left(-\frac{1}{T} \sum_t (\mathrm{Y}_t - \beta X_t)^2 - \lambda \sum_i |\beta_i| \right), \tag{A.4}$$

$$\Longleftrightarrow \underset{\beta}{\operatorname{argmax}} \left[\frac{T}{2\hat{\sigma}^2} \left(-\frac{1}{T} \sum_t (\mathrm{Y}_t - \beta X_t)^2 - \lambda \sum_i |\beta_i| \right) \right].$$

$$\Longleftrightarrow \underset{\beta}{\operatorname{argmax}} -\frac{1}{2} \frac{\sum_t (\mathrm{Y}_t - \beta X_t)^2}{\hat{\sigma}^2} - \frac{\lambda T}{2\hat{\sigma}^2} \sum_i |\beta_i|.$$

$$\Longleftrightarrow \underset{\beta}{\operatorname{argmax}} \exp\left[-\frac{1}{2} \frac{\sum_t (\mathrm{Y}_t - \beta X_t)^2}{\hat{\sigma}^2} \right] \times \exp\left(-\frac{\lambda T}{2\hat{\sigma}^2} \sum_i |\beta_i| \right).$$

$$\Longleftrightarrow \underset{\beta}{\operatorname{argmax}} \left\{ \prod_t \exp[-\frac{1}{2} \frac{(\mathrm{Y}_t - \beta X_t)^2}{\hat{\sigma}^2}] \right\} \times \left[\prod_i \exp(-\frac{\lambda}{2\hat{\sigma}^2} |\beta_i|) \right]^T.$$

$$\Longleftrightarrow \underset{\beta}{\operatorname{argmax}} \left(\frac{1}{\sqrt{2\pi\hat{\sigma}^2}} \right)^T \left\{ \prod_t \exp\left[-\frac{1}{2} \frac{(\mathrm{Y}_t - \beta X_t)^2}{\hat{\sigma}^2} \right] \right\} \times \left(\frac{\lambda}{4\hat{\sigma}^2} \right)^{NT} \left[\prod_i \exp(-\frac{\lambda}{2\hat{\sigma}^2} |\beta_i|) \right]^T.$$

As a result, we can obtain Eq. (19), which is

$$\hat{\beta} = \underset{\beta}{\operatorname{argmax}} \left\{ \prod_t \frac{1}{\sqrt{2\pi\hat{\sigma}^2}} \exp\left[-\frac{1}{2} \frac{(\mathrm{Y}_t - \beta X_t)^2}{\hat{\sigma}^2} \right] \times \left[\prod_i \frac{\lambda}{4\hat{\sigma}^2} \exp(-\frac{\lambda}{2\hat{\sigma}^2} |\beta_i|) \right]^T \right\}. \tag{A.5}$$

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